

ECE 3300 Reaction Game Final
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The game we made is a reaction game that is started by pushing the center button, both rgb LEDs switch to red as well as the computer screen, there is then a delay based on number switches turned on before the LEDs and screen both switch to green when the player must push the left button to react and stop the timer. After the left button is pressed, the screen switches to blue and the timer is stopped, showing the reaction time in milliseconds on both the computer screen and 7 segment display. Another switch can also be used to check the best reaction time when flipped up while the game has been played and not reset.

For this game we used 9 switches, 3 buttons, 2 rgb LEDs, and a 7-segment display. We used switch 0-8 to set a random delay for the red screen to switch to green. The other switch we used was switch 15 that can be flipped up to check fastest reaction time or kept down to see the live reaction time. The three buttons we used are the center button to start the game, the left button used to react and stop the timer, and the cpu reset button to reset the game. The two rgb are used as a visual cue similar to the computer screen. The 7-segment display is used for our timer counting in milliseconds when the screen turns green waiting for the reaction button to be pressed.

The FSM is used to manage the transition between our four separate game states. Our four game states are IDLE, DELAY, ACTIVE, and DONE. The game starts in IDLE and waits for the game to start before it transitions to the DELAY state where it waits for a certain amount of time based on the number of switches turned on by the user. When the timer is finished the FSM switches to the ACTIVE state, the screen and LED switches to green and starts the timer showing the players reaction time. Once the left button is pressed the game transitions to the DONE state and changes the screen to blue showing the players time in milliseconds, then waits for the player to either start another game or reset the game. Our FIFO acts as a data buffer and stores the reaction time scores in a memory array. This helps our game so that it records a time immediately and gets ready for the next game to start without having to wait for the hardware to finish sending the previous data.

The three modeling styles we used were structural, behavioral, and dataflow modeling. We used structural modeling in our system when we made our multiplexer. We used behavioral modeling in many of our modules such as our FIFO but mostly in our game_FSM and our top-level module. The last modeling style we used was dataflow and was used continuously throughout our system with the use of assign statements such as assign our switches to LEDs.

This strategy of our testbench was to mimic an actual player interacting with the game. We used tasks so that the testbench would start the game on its own and simulate a real player delay reaction after reaching the ACTIVE state. In order for us to maximize our simulation time, we used a 'define SIMULATION in order to shorten our cycles down to 10 so that our entire game can be verified in microseconds of simulation time.

